B.N. MANDAL UNIVERSITY · MADHEPURA · 2-YEAR B.ED PART I

A complete, descriptive reading companion with diagrams

Teaching of *Biological* Science

PAPER PSS-5 · SESSION 2024-26 · ENGLISH NOTES · ALL FIVE UNITS · FULL 8-MARK ANSWERS · EVERY DIAGRAM LABELLED

Time · 3 hours *Full marks · 40* *Answer five, one per Group* *~8 marks each*

This booklet follows the official syllabus exactly. Each *Group* in the question paper corresponds to one *Unit* in the syllabus, so the five units below are your five answer slots. Every previous-year question is written out as a full, descriptive answer of the length an 8-mark question expects, with a labelled diagram wherever the topic allows.

How to write a full 8-mark answer (use this skeleton every time)

i. **Definition / meaning** in one or two clear sentences, with a scholar's name or word-origin if you know one.
ii. **Body** in labelled points or short headings: explain each idea, do not just name it.
iii. **Diagram**, neatly labelled, for any structure, process or cell.
iv. **Example** from biology or the classroom to show application.
v. **Merits and limits** (for methods) or **significance** (for biology topics).
vi. **Conclusion**: one or two lines that tie the answer together.

Where each topic lives

The syllabus at a glance

UNIT / GROUP	NATURE	CORE TOPICS
I	Pedagogy	Nature & philosophy of science, aims & objectives, behavioural objectives, lesson & unit planning, teaching aids, constructivism / NCF-2005
II	Pedagogy	Methods of teaching (lecture, discussion, demonstration, heuristic, project, problem solving, inductive-deductive, programmed), evaluation, test items, achievement & diagnostic tests
III	Biology	Nutrition & balanced diet, deficiency diseases, animal tissues, communicable / non-communicable diseases, cell structure, mitosis & meiosis, meristems, tissue systems, permanent & epidermal tissue
IV	Biology	Prokaryotic & eukaryotic cells, organelle functions, cell division, meristems, permanent tissue, epidermis
V	Biology	Feeding mechanism & nutrients, balanced diet & deficiency, diseases, epithelial / connective / muscular / nervous tissue, NCF-2005 activity-based learning

Contents

I

Group I · Pedagogy

Nature of Science, Objectives & *Lesson Planning*

HISTORY & PHILOSOPHY | OBJECTIVES | LESSON & UNIT PLAN | TEACHING AIDS | NCF-2005

1.1 What is Science? Meaning and characteristics

The word *Science* comes from the Latin *scientia*, meaning "to know". Before the eighteenth century science was called *natural philosophy*. Science is not only observation, experiment and analysis; it is a way of life and an ever-expanding body of knowledge built through a continuous process of inquiry. The truest definition is practical: the way to learn science is to *do* science.

TWO USEFUL DEFINITIONS

Fitzpatrick: "Science is a cumulative and endless series of empirical observations which result in the formation of concepts and theories, both being subject to modification in the light of further empirical observations."

Einstein: Science is the attempt to make the chaotic diversity of our sense experience correspond to a logically uniform system of thought.

Important characteristics of science

- It deals with the **natural world**, not supernatural myths (for example, the growth of plants, the characteristics of animals).
- It is not merely a collection of facts; it explains happenings through **analysis, testing and verification**.
- It is **empirical**: every claim must rest on observation or experiment.
- It is **dynamic and tentative**: knowledge is long lasting yet open to change.
- It is **objective** and removes personal bias.
- It is **systematic, self-correcting** (peer review, replication) and a **creative** human endeavour.

– Science and technology are **not the same**, but each affects the other.

PYQ 2025 · Q1 8 MARKS

"Describe the philosophy and nature of Science."

Introduction. Science is at once a body of knowledge, a method of inquiry and a way of thinking about the world. To answer this question fully we examine two related ideas: the *nature* of science (what science is and how it behaves) and the *philosophy* of science (the reasoning and values that lie behind it).

A. The nature of science — three faces

i. **Science as a systematic body of knowledge.** It is an organised store of facts, concepts, principles, laws and theories. This store is not fixed: it is subject to error and change, and is constantly rearranged and reorganised in the light of new evidence. Science is therefore dynamic, not static.

ii. **Science as a way of investigating.** Scientists approach any problem in an organised manner through the *scientific method*: observation, asking a question, forming a hypothesis, experimenting, collecting and analysing data, and drawing a conclusion. Imagination, insight and creativity are part of this process.

iii. **Science as a way of thinking.** This produces the *scientific attitude*, marked by open-mindedness, objectivity, freedom from superstition, belief in cause and effect, accuracy and truthfulness in reporting, a methodical approach to problems, and up-to-dateness.

Key aspects of the nature of science

- Scientific knowledge is long lasting yet tentative.
- Empirical evidence is used to refine and support ideas.
- Social and historical factors shape the construction of scientific knowledge.
- Laws and theories both matter, but serve different functions.
- Accurate record keeping, peer review and replication validate findings.
- Science is creative; science and technology influence each other yet remain distinct.

B. The philosophy of science

The philosophy of science studies how scientific knowledge is created and which values guide it. Its driving force is **scepticism**: every new claim is met with healthy suspicion and accepted only after independent checking yields the same result. Karl Popper added that a statement is scientific only if it is **falsifiable**, that is, capable of being tested and possibly proved wrong. Science rests on at least four fundamental values:

- Curiosity is good and should be encouraged.
- Knowledge itself is good; it is good to acquire it.
- It is wrong to falsify or fabricate the data on which knowledge is based.
- It is good to keep an open mind, tempered by a vigilant scepticism.

Conclusion. The nature of science shows it to be dynamic, evidence-based, self-correcting and creative; its philosophy shows honesty, curiosity, open-mindedness, scepticism and testability to be its guiding values. A good science teacher must pass on both the *knowledge* of science and this *way of thinking*.

1.2 Meaning, nature and scope of Biological Science

Biology (Greek *bios* = life, *logos* = study) literally means "the study of life". It is the natural science of living organisms: their structure, function, growth, evolution, distribution, identification and classification. Its scale runs from the chemical machinery inside a single cell to whole ecosystems and global climate.

- **Nature:** biology is a social endeavour in which knowledge becomes power and wisdom; it can be beneficial or harmful, and in a progressive society it plays a liberating role, freeing people from ignorance and superstition.
- **Scope:** botany, zoology, microbiology, genetics, ecology, physiology, biotechnology, and the applied fields of medicine, agriculture and environmental science.

1.3 Importance of science in daily life

- **Health and medicine:** vaccines, hygiene, balanced nutrition, disease control.
- **Food and agriculture:** improved crops, fertilisers, food preservation.
- **Daily comfort and communication:** electricity, transport, phones, the internet.
- **Environment:** understanding pollution, conservation and sustainable living.
- **Attitude:** it builds rational, evidence-based thinking and removes blind superstition.

1.4 Aims and objectives of teaching biological science

Aims are broad, long-term goals; **objectives** are specific and achievable within a lesson or two.

Aims

- Develop responsible democratic citizens.
- Show how science and technology develop and affect society.
- Encourage the responsible use of science and technology.
- Develop critical thinking about science, technology and society.
- Build a scientific attitude and scientific vision.
- Foster care for a healthy, sustainable environment.

Objectives (Bloom's domains)

- **Cognitive:** knowledge, comprehension, application, analysis, synthesis, evaluation.
- **Affective:** interest, attitudes, values, appreciation of living things.
- **Psychomotor:** observation, drawing, dissection, handling apparatus, experimentation.

1.5 Behavioural (instructional) objectives

BEHAVIOURAL OBJECTIVE

An objective written in terms of the observable behaviour a learner will show after instruction, using action verbs rather than vague words like "know" or "understand".

Format: Condition + Action verb + Content + Standard.
Example: "After the lesson, the student will be able to **label** at least five parts of a prokaryotic cell on a diagram."

Action verbs by Bloom's level

LEVEL	VERBS TO USE
Knowledge	define, list, name, state, recall
Comprehension	explain, describe, summarise
Application	solve, demonstrate, use, illustrate
Analysis	compare, differentiate, classify
Synthesis	design, construct, prepare
Evaluation	judge, justify, criticise

Steps after objectives: identify the *teaching points* (the small units of content to be taught), *organise the content* in a logical order, and *design learning experiences* (activities, demonstrations, questions) that let pupils reach each objective.

PYQ 2025 · Q2 | 8 MARKS

"Describe the different steps of a Lesson Plan."

LESSON PLAN

A written description of the teaching and learning activities a teacher will carry out in a single class period to achieve specific objectives: the teacher's blueprint for one lesson.

Need and importance. A lesson plan gives the teacher direction and confidence, ensures the right use of time and teaching aids, keeps the lesson organised and goal-directed, prevents the omission of important points, and provides a record for later evaluation and improvement.

The most widely used model is the **Herbartian Five Steps** (J.F. Herbart and his followers). Every lesson plan first carries this preliminary information:

- General information: school, class, subject, topic, date, period, duration.
- General and specific (behavioural) objectives.
- Teaching aids and instructional materials.
- Previous knowledge assumed of the pupils.

The steps of the lesson plan

i. **Introduction / Motivation (Preparation).** The teacher tests previous knowledge with a few questions, creates interest, and then announces the *statement of the aim* (today's topic), linking the new learning to what the pupils already know.
ii. **Presentation.** The heart of the lesson: new content is taught step by step through teaching points, developing questions, explanation, demonstration and teaching aids, with teacher and pupils building the lesson together (often shown in two columns, Teaching Points and Teacher-Pupil Activity).
iii. **Comparison and Association.** New facts are compared and linked with related, already-known facts and examples, so that similarities and differences stand out.
iv. **Generalisation.** From the facts presented, the class arrives at a general rule, principle or definition, drawn from the pupils as far as possible rather than dictated.
v. **Application.** The generalisation is applied to new situations or problems, making the knowledge functional and lasting.
vi. **Recapitulation / Evaluation.** Review questions check whether the objectives were achieved, errors are corrected, and a home assignment is given.

Example (Biology): for the topic "Mitosis", the aim is stated, the stages are presented one by one with a chart, mitosis is compared with meiosis, the class generalises that mitosis keeps the chromosome number constant, applies this to growth and wound healing, and finally recaps with questions.

Note: a unit plan is broader, covering a whole topic over several periods; a lesson plan covers a single period within that unit.

1.6 Instructional materials and teaching aids

Teaching aids make teaching clear, effective and interesting. They follow a simple rule: the more senses involved, the better the learning (Edgar Dale's "cone of experience").

Audio

Radio, audio recordings, tape recorder.

Visual

Chart, model, blackboard, picture, specimen, OHP, improvised apparatus, herbarium, aquarium.

Audio-visual

Television, film, computer, multimedia, LCD projector, video.

Components of instructional materials: multimedia, computer, charts, models, improvised apparatus, and prepared unit and lesson plans. A good aid is simple, accurate, large enough to be seen, relevant, low in cost and easy to handle. **Improvised apparatus** is made from cheap local material (bottles, straws, cardboard) when standard equipment is unavailable, and is central to activity-based learning.

1.7 Constructivism and NCF-2005

CONSTRUCTIVISM

The view that learners actively build their own knowledge by connecting new experiences to prior knowledge, rather than receiving facts passively. The teacher becomes a facilitator.

NCF-2005 strongly recommends this approach: learning by doing, activity-based teaching, and connecting knowledge to life outside school. The full account of NCF-2005 is given in Unit V (section 5.6).

II

Group II · Pedagogy

Methods of Teaching & *Evaluation*

HEURISTIC & PROJECT

INDUCTIVE-DEDUCTIVE

PROGRAMMED INSTRUCTION

ACHIEVEMENT & DIAGNOSTIC TESTS

2.1 The methods at a glance

Methods named in the syllabus

METHOD	CORE IDEA	TEACHER ROLE	BEST FOR
Lecture	Teacher tells, pupils listen	Active speaker	Large content, theory
Discussion	Group exchange of ideas	Moderator	Open topics, attitudes
Demonstration	Teacher performs, pupils observe	Performer	Experiments, processes
Heuristic	Pupil discovers for himself	Guide only	Scientific attitude
Project	Learning by doing a real task	Facilitator	Application, integration
Problem solving	Solve a problem scientifically	Guide	Reasoning
Inductive	Examples first, then rule	Questioner	Building concepts
Deductive	Rule first, then examples	Explainer	Quick application
Programmed	Self-learning in small steps	Material designer	Individual learning

PYQ 2025 · Q3 8 MARKS

"Describe the Heuristic method of teaching Biological Science."

Meaning. *Heuristic* comes from the Greek *heurisko*, meaning "I find out" or "I discover". In this method the learner is placed in the position of a discoverer and learns by finding things out for himself instead of being told. It was developed and popularised by **Prof. Henry Edward Armstrong**, who described it as a method that "puts the pupil in the position of a discoverer".

Central principle. The way to learn science is to do science. The student is made to act like a small scientist who observes, experiments, reasons and concludes on his own, so that he learns the *process* of science and not only its results.

Role of the teacher and pupil. The teacher is only a guide and friend: he sets the problem, supplies materials and asks leading questions, but never gives the answer. The pupil is active throughout, working, recording and reasoning.

Steps in the heuristic method

i. The teacher presents a clear problem.
ii. Students are given the materials and apparatus needed.
iii. Students observe and experiment to gather facts.
iv. Students record their observations carefully.
v. Students reason out and generalise from their own data.
vi. The conclusion is verified and then applied to new situations.

Example in biology

Instead of telling students that leaves release oxygen, the teacher gives them Hydrilla, a beaker, water and a funnel and asks them to find out which gas is released in sunlight. The students set up the experiment, collect and test the gas with a glowing splinter, and discover for themselves that oxygen is given out during photosynthesis.

Merits

- Develops scientific attitude and the habit of inquiry.
- Knowledge is permanent because it is self-discovered.
- Builds self-reliance, observation and reasoning power.
- Makes learning active, joyful and interesting.
- Trains pupils directly in the scientific method.

Demerits

- Very slow; only a little content is covered.
- Unsuitable for young or weak pupils who need guidance.
- Needs a highly skilled teacher and a well-equipped laboratory.
- Impractical for large classes or the whole syllabus.
- Expects every child to be a discoverer, which is not realistic.

Conclusion. The heuristic method is excellent for building scientific attitude and process skills, but because it is slow and demanding it is best combined with other methods rather than used alone.

2.2 The methods of teaching, described in full

A **teaching method** is the planned way in which a teacher presents content so that pupils learn. The heuristic method is given in the answer above; the other methods named in the syllabus are described below, each with its meaning, steps, merits and demerits. This is the material for any "describe the method of teaching" question.

1 · Lecture method

Meaning: the oldest and most teacher-centred method, in which the teacher presents facts, principles and information orally while the pupils listen and take notes.

Steps: preparation of the matter, introduction to rouse interest, presentation by oral exposition, and recapitulation with questions.

Merits: covers much content quickly, economical, suits large classes, good for theory and background. **Demerits:** pupils stay passive, no activity or "doing", ignores individual differences, poor for skills, encourages rote learning.

2 · Demonstration (lecture-cum-demonstration) method

Meaning: the teacher performs an experiment or shows a specimen, model or chart while explaining it, so that pupils learn by both seeing and hearing ("to demonstrate" means "to show").

Steps: planning the demonstration, introduction, performing it step by step with developing questions, observation by pupils, generalisation, and recapitulation.

Merits: joins the spoken word with a visual experience, is useful when apparatus is costly, risky or few, saves time and material, and develops the power of observation. **Demerits:** pupils mostly watch instead of doing, back-benchers may not see clearly, and there is no individual handling of apparatus.

3 · Discussion method

Meaning: a cooperative exchange of ideas between the teacher and pupils, or among the pupils themselves, on a chosen problem or topic. Forms include whole-class discussion, group discussion, panel and symposium.

Steps: select and announce the topic, plan the points, conduct the discussion with the teacher as moderator, and summarise the conclusions.

Merits: active participation, develops expression, reasoning and tolerance of other views, and builds healthy attitudes. **Demerits:** time-consuming, may wander off the point, can be dominated by a few pupils, and is weak for learning plain facts.

4 · Inductive and Deductive methods

Inductive: teaching moves from particular examples and observations to a general rule, formula or law (from the specific to the general). Steps: present examples, observe, compare, and generalise. It gives real understanding but is slow.

Deductive: teaching moves from the general rule to particular examples (from the general to the specific). Steps: state the rule, then apply it to examples. It is quick and short but can become rote. In practice the two are used together: induction to discover a rule, deduction to apply it.

5 · Project method

Meaning: learning by doing a purposeful, real-life task in a natural setting, based on Dewey's and Kilpatrick's "learning by doing". Kilpatrick called a project "a wholehearted purposeful activity proceeding in a social environment".

Steps: (i) providing a situation, (ii) choosing the project, (iii) planning, (iv) executing or carrying it out, (v) evaluating, and (vi) recording. Examples: a school kitchen garden, a herbarium, a vermicompost pit, or a model of a cell.

Merits: learning by doing, joyful and motivating, develops cooperation and dignity of labour, and links school to real life. **Demerits:** time-consuming and costly, the syllabus is not covered in order, and progress is hard to assess.

6 · Problem-solving method

Meaning: pupils are faced with a real problem and solve it through the steps of the scientific method, so that they learn to reason for themselves.

Steps: (i) recognising the problem, (ii) defining it clearly, (iii) forming hypotheses, (iv) collecting and organising data, (v) testing the hypotheses and drawing a conclusion, and (vi) applying it.

Merits: develops reasoning, scientific attitude and self-confidence, and the learning is lasting. **Demerits:** slow, not suited to every topic, and it needs able pupils and a skilled teacher.

7 · Programmed instruction (programmed learning)

Meaning: an individualised, self-instructional technique in which the content is broken into small steps called frames; the learner responds at each step and gets immediate feedback. It is based on Skinner's operant conditioning.

Principles: small steps, active responding, immediate confirmation, self-pacing and self-testing. **Types:** Linear (Skinner) and Branching (Crowder).

Merits: each pupil learns at his own pace with instant reinforcement, and errors are reduced. **Demerits:** costly to prepare, lacks the human and emotional touch, can be monotonous, and is poor for skills and attitudes.

8 · Inquiry / Activity-based learning (NCF-2005 constructivist approach)

Meaning: pupils learn by actively investigating questions and carrying out activities, building (constructing) their own knowledge, while the teacher acts as a facilitator. NCF-2005 recommends this for science.

Features: child-centred, learning by doing, use of real materials, group work, and a close link with the pupil's environment.

Merits: deep understanding, scientific process skills and high motivation. **Demerits:** needs much time, resources and trained teachers.

2.3 Evaluation and types of test items

EVALUATION

The process of judging how far the learning objectives have been achieved. It may be oral or written, and formative (during teaching) or summative (at the end).

Types of questions

- **Essay type:** long answers; test expression and organisation, but subjective to mark.
- **Short answer:** a few sentences; cover more content with more objectivity.
- **Objective:** one correct answer; fully objective. Forms: MCQ, fill in the blanks, true-false, matching.

Types of tests

- **Achievement test:** measures how much was learnt after teaching.
- **Diagnostic test:** finds the specific weaknesses of a pupil so that remedial teaching can be planned.

Types of test items, with an example

ITEM TYPE	WHAT IT IS	EXAMPLE
Multiple choice (MCQ)	A stem followed by four options, one correct	The powerhouse of the cell is the (a) ribosome (b) mitochondrion (c) nucleus (d) vacuole
Fill in the blank	A sentence with a key word left out	Scurvy is caused by lack of vitamin ______
True or false	A statement to be judged correct or wrong	Cardiac muscle is voluntary. (True / False)
Matching	Two columns to be paired correctly	Xylem ↔ conducts water; Phloem ↔ conducts food
Short answer	Answered in a few sentences	Define a balanced diet.
Essay (long answer)	A long, organised answer	Describe the stages of mitosis with a diagram.

Making good test items: the language should be simple and clear, each item should test one point, there must be only one correct answer for objective items, and the items should match the objectives and content of the blueprint.

PYQ 2025 · Q4 | 8 MARKS

"Prepare a Biology achievement test of 100 marks including objective and short-answer type questions."

An **achievement test** measures how much a pupil has learnt after a course of instruction. A good test has four qualities: **validity** (it tests what it claims to), **reliability** (it gives consistent results), **objectivity** (marking is free of bias) and **usability** (it is practical to give and score). It is prepared in three stages: **Planning (the blueprint), Preparation (item writing), and Try-out and scoring**.

Step 1 — Planning (the blueprint)

A blueprint gives weightage to objectives, to content and to the form of questions.

Weightage to objectives

OBJECTIVE	MARKS
Knowledge	35
Understanding	30
Application	20
Skill	15
Total	100

Weightage to form of questions

TYPE	NO.	EACH
Objective	40	1
Short answer	12	3
Essay	3	8
Total	55	

Weightage to content

UNIT	MARKS
The cell and cell division	25
Tissues (plant and animal)	25
Nutrition and balanced diet	25
Diseases and health	25

Step 2 — Preparation (a sample of items)

Section A · Objective (40 marks, 1 each)

- *MCQ:* The powerhouse of the cell is the (a) ribosome (b) mitochondrion (c) nucleus (d) vacuole.
- *MCQ:* Scurvy is caused by deficiency of vitamin (a) A (b) B (c) C (d) D.
- *Fill up:* The hereditary material of the cell is __________.
- *True/False:* Cardiac muscle is voluntary.
- *Matching:* xylem, phloem, neuron, goitre ↔ conducts water, conducts food, conducts impulse, iodine deficiency.

Section B · Short answer (36 marks, 3 each)

- Define a balanced diet.
- Differentiate mitosis and meiosis (any three points).
- Name two communicable and two non-communicable diseases.
- State three functions of xylem.

Section C · Essay (24 marks, 8 each, answer any three)

- Describe the stages of mitosis with a diagram.
- Explain the structure and function of nervous tissue.
- What is a balanced diet? Explain any two deficiency diseases.

Step 3 — Try-out and scoring. Prepare a marking scheme with model answers, give the test, score it, and (for a standardised test) carry out item analysis to find the difficulty and discrimination of each item before final use. In the exam, write the three blueprint tables plus a sample of each item type, as above.

III

Group III · Biology

Nutrition, Cell Division & *Disease*

MITOSIS & MEIOSIS | BALANCED DIET | DEFICIENCY DISEASES | TISSUE SYSTEMS

PYQ 2025 · Q5 | 8 MARKS

"What is Mitosis? Describe a short account of its various stages."

A type of cell division in which one parent cell divides into two daughter cells, each with the same chromosome number and identical genetic material as the parent. It occurs in somatic (body) cells and is called equational division.

Before mitosis: Interphase. The cell first passes through interphase, made of three phases: *G1* (cell grows and makes proteins), *S* (DNA replicates, so each chromosome now has two sister chromatids) and *G2* (final growth and preparation). Interphase is preparation; the actual division has two parts, **karyokinesis** (division of the nucleus, in four stages, P-M-A-T) and **cytokinesis** (division of the cytoplasm).

The four stages of mitosis

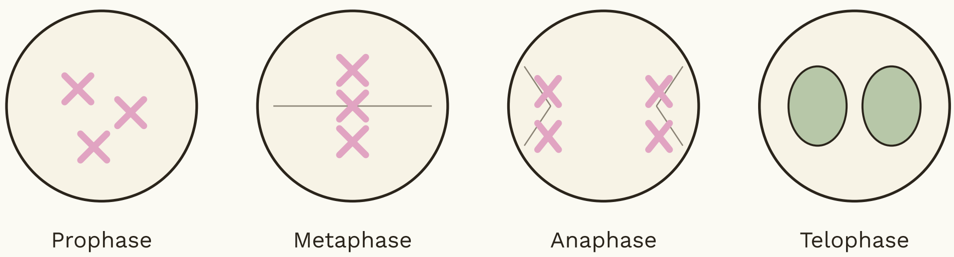
i. **Prophase.** Chromatin condenses into distinct chromosomes, each of two sister chromatids joined at the centromere. The nuclear membrane and nucleolus begin to disappear, the centrioles move to opposite poles, and the spindle fibres begin to form.

ii. **Metaphase.** The chromosomes line up at the equator of the cell (the metaphase plate), and the spindle fibres attach to the centromere of each chromosome. Chromosomes are most clearly visible at this stage.

iii. **Anaphase.** The centromeres split, the two sister chromatids of each chromosome separate, and the daughter chromosomes are pulled to opposite poles by the spindle fibres.

iv. **Telophase.** The chromosomes reach the poles and uncoil back into chromatin, the nuclear membrane and nucleolus reappear around each group, and the spindle fibres disappear. Two nuclei are now formed.

Cytokinesis. The cytoplasm divides: in animal cells by a *cleavage furrow* that pinches the cell in two, and in plant cells by a *cell plate* that grows into a new cell wall. The result is two identical diploid daughter cells.



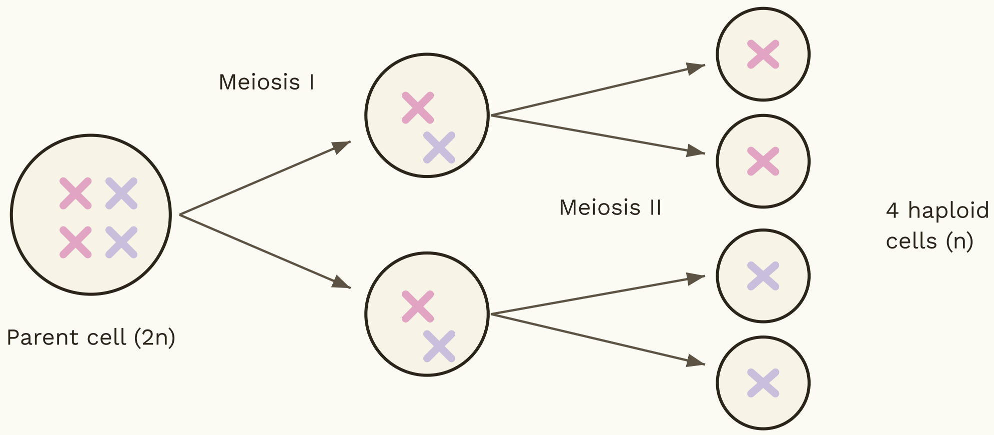
The four stages of mitosis (karyokinesis): chromosomes condense and scatter, line up at the equator, separate to opposite poles, then re-form two nuclei.

Significance of mitosis: it brings about growth of the body, repair and healing of wounds, replacement of dead cells, and asexual reproduction in some organisms, while keeping the chromosome number constant from one cell generation to the next.

If you draw it by hand: four circles showing chromosomes scattered, lined at the centre, pulled to two sides, then two nuclei forming.

3.1 Meiosis (for comparison)

Meiosis is the division of reproductive (germ) cells, producing four daughter cells each with half (haploid) the chromosome number. It is called reductional division and has two divisions, Meiosis I and II. In Meiosis I, homologous chromosomes pair and **crossing over** occurs (exchange of segments, the source of variation), after which the homologues separate; Meiosis II resembles mitosis. Meiosis keeps the chromosome number constant across generations (since fertilisation later doubles it) and creates genetic variation.



Meiosis: one diploid parent cell undergoes two divisions to give four haploid cells, with crossing over giving variation.

Mitosis versus meiosis

FEATURE	MITOSIS	MEIOSIS
Occurs in	Body cells	Germ cells
Divisions	One	Two
Daughter cells	2	4
Chromosome number	Same (diploid)	Half (haploid)
Crossing over	Absent	Present
Purpose	Growth, repair	Gametes, variation

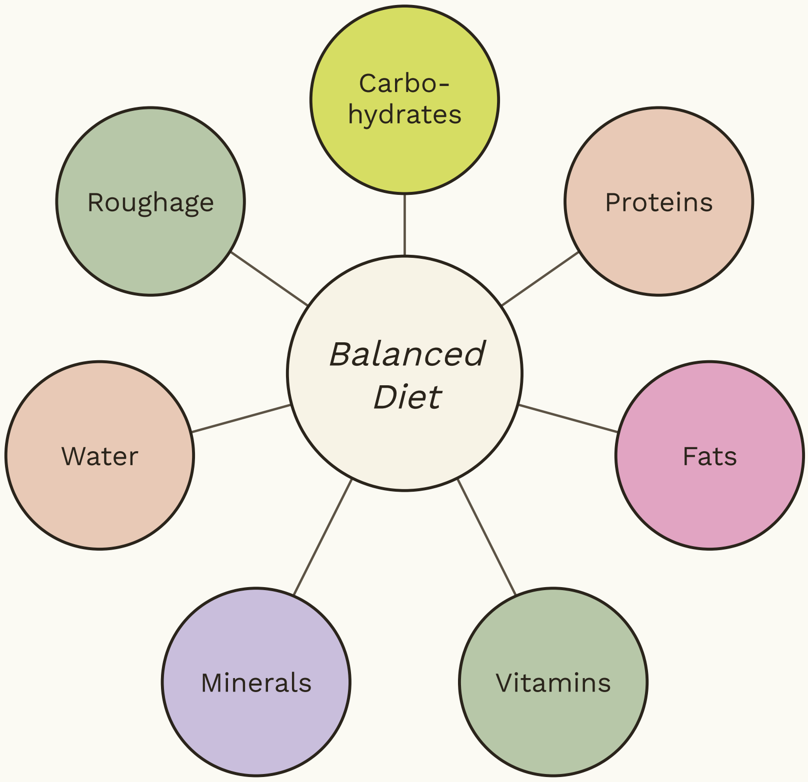
PYQ 2025 · Q6 8 MARKS

"What do you mean by Balanced Diet? Explain any two diseases caused by nutritional deficiency."

BALANCED DIET

A diet that contains all the essential nutrients (carbohydrates, proteins, fats, vitamins, minerals, water and roughage) in the correct proportion and amount needed by the body for growth, energy, maintenance and good health.

The exact balance depends on a person's **age, sex, body weight, kind of work and climate**. A growing child and a labourer need more energy and protein than an old or sedentary person. A diet lacking even one nutrient over a long time leads to *malnutrition* and a *deficiency disease*.



The seven components of a balanced diet.

Components and their functions

NUTRIENT	MAIN FUNCTION	SOURCES
Carbohydrates	Provide energy	Rice, wheat, potato, sugar
Proteins	Growth and repair (body building)	Pulses, milk, egg, fish
Fats	Concentrated energy, store	Oil, ghee, butter, nuts
Vitamins	Protective, regulate body functions	Fruits, vegetables
Minerals	Build bone and blood; regulate	Milk, greens, salt
Water	Medium for reactions, transport	Water, fruits
Roughage	Helps digestion, clears bowels	Salad, whole grain

Two deficiency diseases explained

1 · Scurvy (Vitamin C deficiency)

- **Cause:** long-term lack of Vitamin C (ascorbic acid), found in citrus fruit, amla, tomato and green vegetables.
- **Symptoms:** bleeding and swollen gums, loosening and falling of teeth, slow healing of wounds, weakness, joint pain, anaemia.
- **Prevention / cure:** fresh citrus fruits and amla in the diet.

2 · Rickets (Vitamin D deficiency)

- **Cause:** lack of Vitamin D and calcium; Vitamin D is made in the skin in sunlight and is found in milk, egg yolk, fish and butter.
- **Symptoms:** soft and weak bones, bent or bow legs, deformed chest, delayed teeth in children. In adults the same lack causes osteomalacia.
- **Prevention / cure:** morning sunlight, and milk, eggs and fish in the diet.

Other deficiency diseases (keep this table ready)

DEFICIENT NUTRIENT	DISEASE
Protein (children)	Kwashiorkor, Marasmus
Vitamin A	Night blindness, Xerophthalmia
Vitamin B1 (thiamine)	Beriberi
Vitamin B3 (niacin)	Pellagra
Vitamin B12 / Iron	Anaemia
Iodine	Goitre
Calcium / Vitamin D	Rickets, Osteomalacia
Vitamin C	Scurvy

3.2 Communicable and non-communicable diseases

Communicable (infectious) diseases are caused by germs (pathogens such as bacteria, viruses, protozoa, fungi and worms) and spread from a sick person to a healthy one through air, water, food, direct contact or carriers (vectors) like the mosquito and the housefly. **Non-communicable diseases** do not spread from person to person; they are caused by faulty lifestyle, a lack of nutrients, heredity or the failure of an organ.

Common communicable diseases: cause, spread, symptoms, prevention

DISEASE	PATHOGEN	SPREAD BY	SYMPTOMS	PREVENTION / CONTROL
Tuberculosis (TB)	Bacterium	Air (droplets)	Long cough, fever, weight loss	BCG vaccine, isolation, hygiene
Cholera	Bacterium	Dirty water / food	Watery diarrhoea, vomiting	Safe water, sanitation, ORS
Typhoid	Bacterium	Contaminated food / water	Long fever, weakness	Hygiene, vaccine, clean water
Malaria	Protozoan (Plasmodium)	Female Anopheles mosquito	Fever with chills, sweating	Kill mosquitoes, nets, drain water
Influenza / common cold	Virus	Air (droplets)	Fever, cough, running nose	Rest, hygiene, avoid contact
Hepatitis / jaundice	Virus	Water, food, blood	Yellow eyes and skin, fever	Vaccine, safe water and blood
Ringworm	Fungus	Contact, shared items	Itchy ring-shaped skin patches	Cleanliness, antifungal cream

Common non-communicable diseases

- **Diabetes:** high blood sugar from a lack of insulin or a faulty lifestyle.
- **Hypertension:** high blood pressure from stress, salt and obesity.
- **Heart disease:** from fatty food, smoking and no exercise.
- **Cancer:** uncontrolled growth of cells.
- **Deficiency diseases:** scurvy, rickets, goitre and the like.

General prevention and control

- Cleanliness, safe drinking water and proper sanitation.
- Vaccination and timely treatment.
- Balanced diet and regular exercise.
- Avoiding tobacco, alcohol and stress.
- Killing vectors and isolating patients.
- Health education and regular check-ups.

3.3 The plant tissue system

The body of a flowering plant is organised into three tissue systems: the **epidermal system** (the outer protective layer, with cuticle and stomata), the **ground tissue system** (parenchyma, collenchyma and sclerenchyma, which fill, store and support), and the **vascular system** (xylem and phloem, which conduct water and food). Apical meristems at the root and shoot tips supply the new cells. The animal tissues named in this unit are treated in full under Unit V.

IV

Group IV · Biology

The Cell & *Plant Tissues*

PROKARYOTIC CELL · ORGANELLES · MERISTEMS · PERMANENT TISSUES

4.1 Prokaryotic and eukaryotic cells

The cell is the basic structural and functional unit of life. It was discovered by **Robert Hooke** in 1665 in a thin slice of cork. The **Cell Theory** (given by Schleiden and Schwann, and completed by Virchow) states that all living things are made of cells, that the cell is the basic unit of life, and that all cells arise from pre-existing cells. Cells are of two kinds, prokaryotic and eukaryotic.

Prokaryotic versus eukaryotic

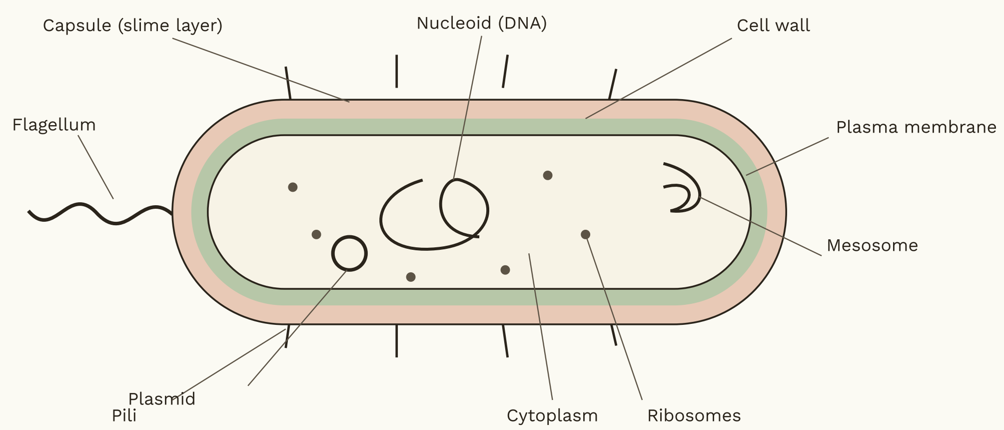
FEATURE	PROKARYOTIC	EUKARYOTIC
Nucleus	No true nucleus (nucleoid)	True, membrane-bound nucleus
Size	Small (1–10 μm)	Larger (10–100 μm)
Membrane-bound organelles	Absent	Present
Ribosomes	70S	80S
DNA	Single, circular, naked	Linear, with proteins
Examples	Bacteria, blue-green algae	Plant, animal, fungal cells

PYQ 2025 · Q7

8 MARKS

"Draw a labelled diagram of a Prokaryotic cell."

Meaning. A prokaryotic cell (example: a bacterium such as *E. coli*) is a cell that has *no true nucleus* and *no membrane-bound organelles*. Its genetic material lies free in the cytoplasm in a region called the nucleoid. Below is the labelled diagram followed by the function of each part.

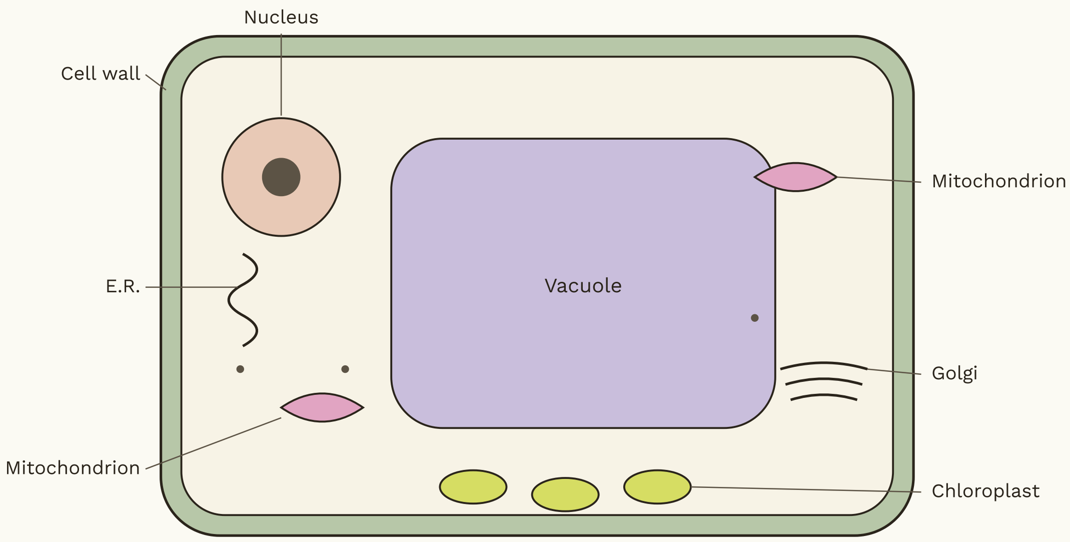


Labelled structure of a prokaryotic (bacterial) cell.

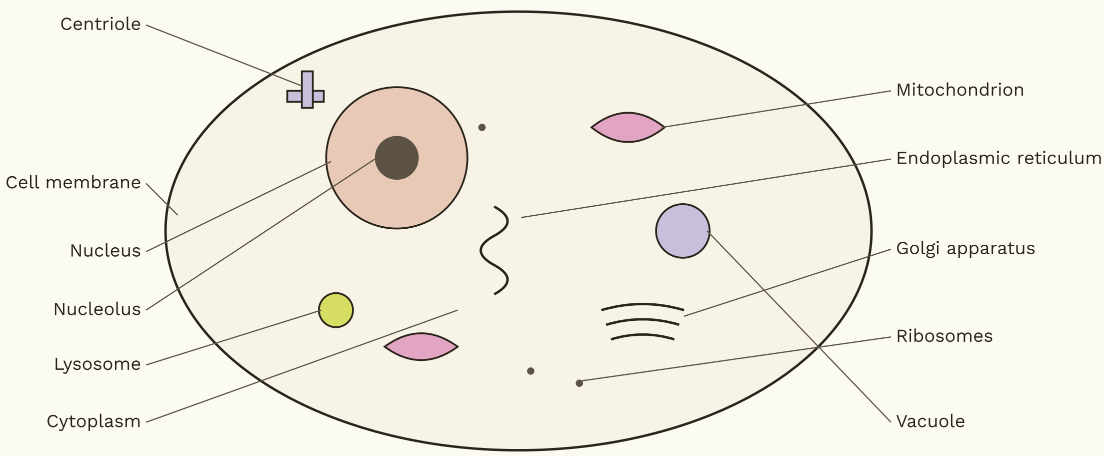
Parts and their functions

PART	FUNCTION
Cell wall	Rigid outer cover (of peptidoglycan); gives shape and protection
Capsule / slime layer	Extra sticky outer coat for protection and attachment
Plasma membrane	Controls the entry and exit of substances
Cytoplasm	Jelly-like fluid where the reactions of the cell occur
Nucleoid	Region holding the single, circular, naked DNA
Plasmid	Small extra circular DNA carrying extra genes
Ribosomes (70S)	Sites of protein synthesis
Mesosome	Infolding of the membrane that aids respiration and division
Flagellum	Whip-like structure for movement
Pili / fimbriae	Hair-like structures for attachment

4.2 Functions of cell organelles (eukaryotic)



A generalised plant cell. The cream inner space is the cytoplasm; the outer green band is the cell wall with the cell membrane just inside it.



A generalised animal cell. It has no cell wall and no chloroplast, and its vacuoles are small.

Structure and function of cell organelles

ORGANELLE	STRUCTURE	FUNCTION
Nucleus	Large, spherical body bound by a double, porous nuclear membrane; holds the chromatin (DNA) and the nucleolus	Controls all the activities of the cell and carries the hereditary material
Mitochondria	Rod- or sausage-shaped, double-membraned; the inner membrane is folded into cristae; has its own DNA	Powerhouse of the cell; releases energy (ATP) by respiration
Ribosomes	Tiny granules of RNA and protein, lying free or attached to the rough ER	Sites of protein synthesis
Endoplasmic reticulum	Network of membranous tubes and sheets; rough ER bears ribosomes, smooth ER does not	Transport within the cell; rough ER makes proteins, smooth ER makes fats (lipids)
Golgi apparatus	Stack of flattened membranous sacs (cisternae) with small vesicles at the edges	Modifies, packages and secretes materials made by the cell
Lysosomes	Small, single-membrane sacs filled with strong digestive enzymes	Digest waste, germs and worn-out parts; the cell's "suicide bags"
Chloroplast (plant)	Double-membraned green plastid; chlorophyll lies in stacks called grana inside the stroma	Carries out photosynthesis (makes food)
Vacuole	Fluid-filled sac bound by a membrane (tonoplast); large and central in plant cells	Stores food, water and waste; keeps the plant cell firm (turgid)
Cell membrane	Thin, living, selectively permeable layer of lipid and protein	Controls the entry and exit of substances

ORGANELLE	STRUCTURE	FUNCTION
Cell wall (plant)	Thick, non-living, rigid outer layer of cellulose	Gives shape, rigidity and protection
Centrosome / centriole (animal)	Two short cylindrical centrioles lying near the nucleus	Forms the spindle fibres during cell division

Difference between a plant cell and an animal cell

FEATURE	PLANT CELL	ANIMAL CELL
Cell wall	Present (of cellulose)	Absent
Shape	Fixed, mostly rectangular	Irregular, rounded
Chloroplast	Present	Absent
Vacuole	One large central vacuole	Small or absent
Centriole	Absent (in higher plants)	Present
Mode of nutrition	Autotrophic (makes food)	Heterotrophic (takes food)

PYQ 2025 · Q8 8 MARKS

"Describe the different types of Permanent tissues."

PERMANENT TISSUE

Tissue made of cells that have lost the power of division and taken up a definite shape, size and function. It is formed from meristematic (dividing) tissue and makes up most of the mature plant body.

A · Simple permanent tissue (one type of cell)

Parenchyma

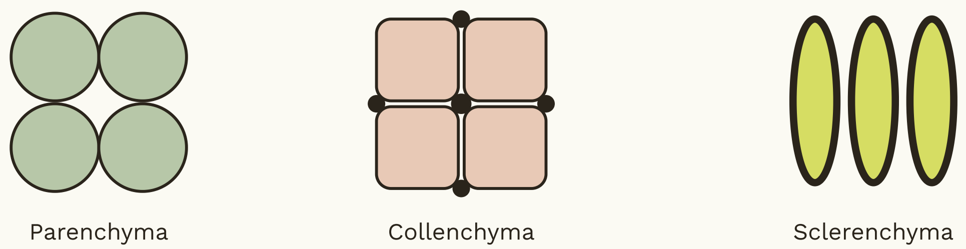
Living, thin-walled cells with spaces between them. They store food and water; when green (chlorenchyma) they photosynthesise; when air-filled (aerenchyma) they give buoyancy to water plants.

Collenchyma

Living cells thickened at the corners. They give flexible mechanical support to young stems and leaf stalks, letting them bend without breaking.

Sclerenchyma

Dead cells with thick, lignified walls and no living content. They give hardness, rigidity and strength (husks, nut shells, fibres such as jute and coir).



Simple permanent tissues: thin-walled parenchyma, corner-thickened collenchyma, and thick-walled dead sclerenchyma.

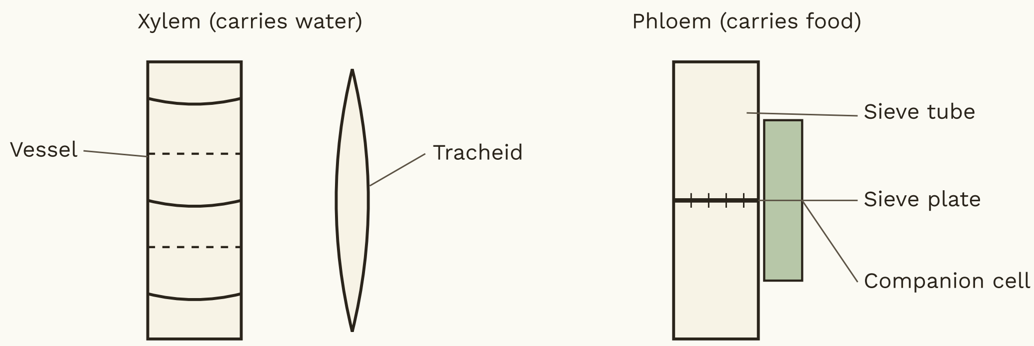
B · Complex permanent tissue (many cell types, conducting)

Xylem

Conducts water and minerals upward from root to leaf and gives mechanical support. Made of four elements: tracheids, vessels, xylem parenchyma and xylem fibres (mostly dead cells).

Phloem

Conducts prepared food from the leaves to all parts (translocation). Made of four elements: sieve tubes, companion cells, phloem parenchyma and phloem fibres (mostly living).



Complex permanent tissues: xylem (vessel and tracheid) conducts water; phloem (sieve tube with sieve plate and companion cell) conducts food.

C · Special / secretory tissue

Some texts add **laticiferous tissue** (contains latex, as in rubber and papaya) and **glandular tissue** (secretes oils, resins and nectar through resin ducts, oil glands and nectaries).

Memory cue: Simple = one cell type (parenchyma, collenchyma, sclerenchyma). Complex = many types (xylem carries water, phloem carries food).

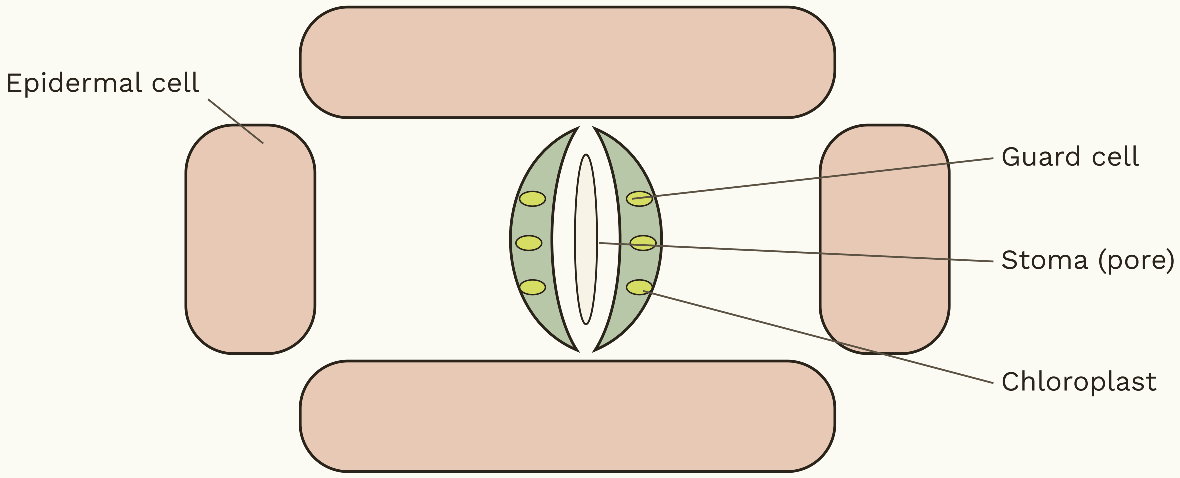
4.3 Meristematic tissue and epidermis

Meristem (dividing tissue)

- **Apical:** at root and shoot tips; increases length.
- **Lateral (cambium):** at the sides; increases girth (thickness).
- **Intercalary:** at the base of leaves or internodes (as in grasses).

Epidermis

The outermost protective layer of the plant, often covered by a waxy cuticle. It guards against water loss, injury and infection; stomata with guard cells in it allow gas exchange.



A stoma: two chloroplast-bearing guard cells enclosing a pore in the epidermis, for gas exchange and transpiration.

V

Group V · Biology

Animal Tissues & *NCF-2005*

FEEDING & NUTRITION | FOUR ANIMAL TISSUES | NERVOUS TISSUE | NCF-2005

5.1 Feeding mechanism and nutrients

Nutrition is the way an organism takes in and uses food. The two broad modes are **autotrophic** (green plants make their own food by photosynthesis) and **heterotrophic** (animals and others take ready-made food). Heterotrophic nutrition is of three kinds: **holozoic** (taking in solid food, as in humans and most animals), **saprophytic** (feeding on dead, decaying matter, as in fungi) and **parasitic** (feeding on a living host, as in tapeworm).

Animals show varied **feeding mechanisms**: *filter feeding* (sieving small particles from water, as in some fish), *fluid feeding* (sucking nectar or blood, as in butterflies and mosquitoes), and *bulk feeding* (swallowing large pieces, as in snakes and humans). The chief **nutrients** are carbohydrates, proteins, fats, vitamins, minerals, water and roughage (detailed under Unit III, with the balanced-diet diagram).

Photosynthesis (how plants make food)

The process by which green plants make their own food (glucose) from carbon dioxide and water, using sunlight trapped by chlorophyll, and release oxygen.

Equation: $6CO_2 + 6H_2O$ → (sunlight, chlorophyll) → $C_6H_{12}O_6 + 6O_2$

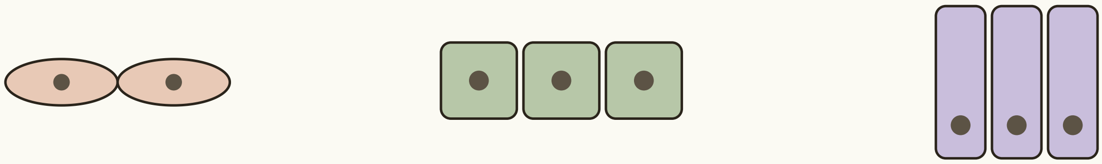
Respiration (how cells get energy)

The breakdown of food (glucose) inside the cell to release energy for life processes; it takes place in the mitochondria.

Equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (ATP)

5.2 The four animal tissues

TISSUE	MAIN JOB	LOCATION
Epithelial	Covering, lining, protection, secretion	Skin, lining of mouth, gut, vessels
Connective	Binding, support, transport	Blood, bone, cartilage, tendon, ligament, fat
Muscular	Movement by contraction	Skeletal, smooth, cardiac muscle
Nervous	Control and coordination	Brain, spinal cord, nerves



Three common shapes of epithelial cells (dark dots are nuclei).

1 · Epithelial tissue

- **Structure:** one or more layers of closely packed cells with almost no matrix between them, resting on a non-living basement membrane; the cells are joined by tight junctions.
- **Types:** squamous (flat, as in the lining of the mouth, blood vessels and air sacs), cuboidal (cube-shaped, in kidney tubules and glands), columnar (tall, lining the stomach and intestine), ciliated (bearing cilia, in the windpipe), glandular (secretes, forming glands) and stratified (many layers, as in the skin).
- **Functions:** protection, absorption, secretion, excretion, filtration, sensation and reduction of friction.

2 · Connective tissue

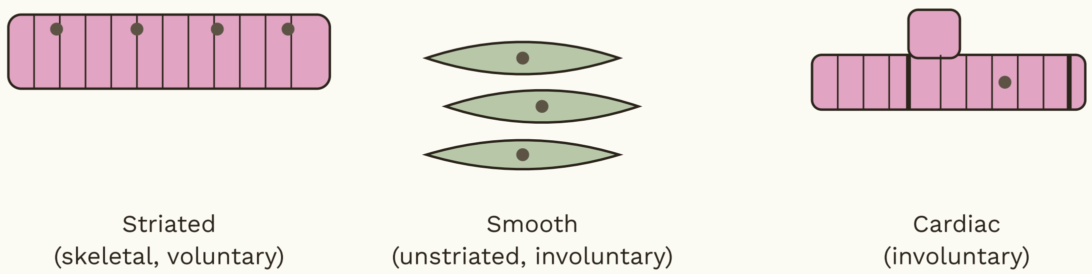
- **Structure:** the most abundant tissue; a few cells lie widely spaced in a large non-living matrix (jelly-like, fluid or solid) that contains fibres such as collagen and elastin.
- **Types:** areolar (packing between organs), adipose (stores fat, insulates), blood (fluid matrix; carries materials), bone (hard matrix of calcium salts; framework), cartilage (flexible; nose, ear, joints), tendon (joins muscle to bone) and ligament (joins bone to bone).
- **Functions:** binds and supports organs, transport (blood), storage (fat), defence (white blood cells) and gives the body its framework.

3 · Muscular tissue

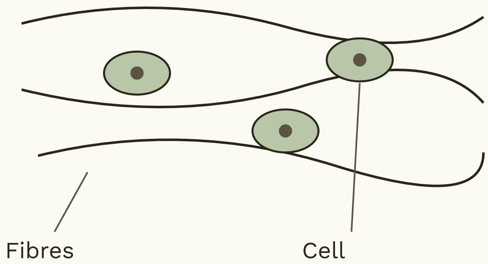
- **Structure:** long, contractile cells called muscle fibres that contain the contractile proteins actin and myosin.
- **Types:** *striated / skeletal* (cylindrical, striped, many nuclei, voluntary, attached to bones), *smooth / unstriated* (spindle-shaped, one nucleus, no stripes, involuntary, in the gut and blood vessels) and *cardiac* (branched, striped but involuntary, in the heart wall, beats rhythmically and never tires).
- **Functions:** movement of the body and its parts, locomotion, movement of food and blood inside the body, posture and the heartbeat.

4 · Nervous tissue

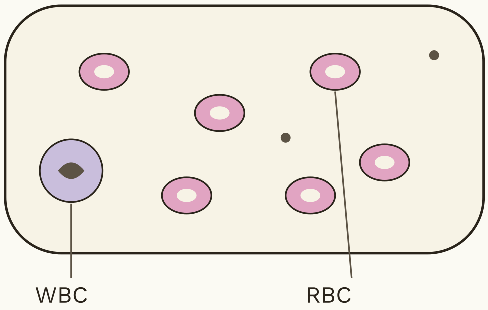
- **Structure:** made of nerve cells (neurons), which carry impulses, and supporting neuroglia cells; the neuron has a cell body, dendrites and an axon (full structure in the answer below).
- **Functions:** receives stimuli, conducts impulses rapidly, and provides control and coordination of all body activities.



The three muscle types: striped voluntary skeletal, spindle-shaped involuntary smooth, and branched involuntary cardiac (with intercalated discs).



Areolar (loose) tissue



Blood

Connective tissue has cells set in a matrix: areolar tissue (cells among fibres) and blood (red cells, white cells and platelets in plasma).

PYQ 2025 · Q9 8 MARKS

"Explain the structure and function of nervous tissue."

NERVOUS TISSUE

The tissue that receives stimuli and conducts impulses from one part of the body to another, providing control and coordination. It builds the brain, spinal cord and nerves.

It is made of two kinds of cells: **neurons** (nerve cells, which carry impulses) and **neuroglia** (supporting cells, which nourish, support and insulate the neurons but do not conduct impulses). The neuron is its structural and functional unit.

Structure of a neuron

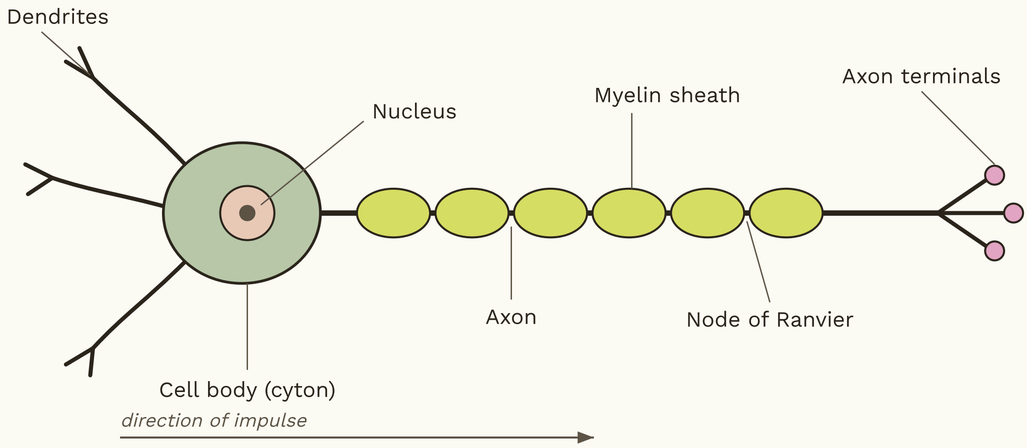
i. **Cell body (cyton or soma):** contains the nucleus and cytoplasm with granules called Nissl granules; it controls the activity of the neuron.

ii. **Dendrites:** short, branched processes that receive impulses and carry them towards the cell body.

iii. **Axon:** a single long fibre that carries the impulse away from the cell body to the next cell.

iv. **Myelin sheath:** a fatty insulating cover around the axon that speeds up the impulse; its gaps are the Nodes of Ranvier, which allow the impulse to jump rapidly.

v. **Axon terminals (synaptic knobs):** the tips of the axon that pass the impulse to the next neuron or to a muscle across a junction called the synapse, using chemicals called neurotransmitters.



Labelled structure of a neuron, the unit of nervous tissue.

Types of neuron

- **Sensory (afferent):** carry impulses from sense organs to the brain or spinal cord.
- **Motor (efferent):** carry impulses from the brain or cord to muscles and glands.
- **Interneuron:** connect sensory and motor neurons.

Functions of nervous tissue

- Receives stimuli from inside and outside the body.
- Conducts nerve impulses rapidly from part to part.
- Coordinates and controls all body activities.
- Produces quick responses, including reflex actions.
- Is the basis of thinking, memory and the senses.

PYQ 2025 · Q10 | 8 MARKS

"Describe the main points of NCF-2005 in light of Science."

The **National Curriculum Framework 2005** was prepared by the NCERT (the steering committee was chaired by Prof. Yash Pal) and grew out of the earlier report "Learning Without Burden" (1993). Its famous theme is to *connect knowledge to life outside the school* and to move from rote (memory) learning towards understanding and learning by doing.

The five guiding principles

i. Connecting knowledge to life outside the school.
ii. Shifting from rote methods of learning to understanding.
iii. Enriching the curriculum so that it goes beyond textbooks.
iv. Making examinations flexible and integrated with classroom life.
v. Nurturing an identity informed by caring concerns within a democratic framework.

NCF-2005 in the light of science

- **Constructivist approach:** the child constructs his own knowledge; the teacher is a facilitator.
- **Learning by doing / activity-based learning** through experiment, observation and projects (the syllabus marks Units III, IV and V for this emphasis).
- **Child-centred** teaching that begins from the child's own experience and environment.
- **Relating science to daily life and the environment.**
- Developing **scientific attitude and process skills**, not only facts.
- **Reducing the curriculum load** (learning without burden).
- Encouraging **inquiry and discovery** (linking to the heuristic and project methods).
- **Examination reform:** continuous and comprehensive evaluation in place of one final memory test.

Six validity criteria for a good science curriculum (NCF-2005)

CRITERION	MEANING
Cognitive validity	Content and method suit the child's age and mental level
Content validity	The content is correct and significant
Process validity	Children learn the method of doing science, not only its conclusions
Historical validity	Science is shown as an evolving, human, social endeavour
Environmental validity	Science is placed in the child's environment and local issues
Ethical validity	It builds honesty, objectivity and concern for life and environment

Conclusion. In the light of science, NCF-2005 wants a classroom that is activity-based, child-centred, constructivist and connected to real life, developing scientific attitude and process skills rather than rote memory.

The night before

Quick *Revision* Sheet

Unit I

- Science = Latin *scientia* = "to know"; natural philosophy before the 18th c.; Fitzpatrick & Einstein definitions.
- Nature = body of knowledge + way of investigating + way of thinking.
- Four values: curiosity good, knowledge good, never falsify data, open mind with scepticism (Popper: falsifiability).
- Lesson plan (Herbart): Introduction, Presentation, Comparison, Generalisation, Application, Recapitulation.

Unit II

- Heuristic = Greek *heurisko* = "I discover"; by **Armstrong**; pupil is discoverer, teacher only guides.
- Achievement test qualities: validity, reliability, objectivity, usability. Steps: blueprint, item writing, scoring.
- Objective items = MCQ, fill-up, true-false, matching. Diagnostic test finds weaknesses.

Unit III

- Mitosis: body cells, 2 identical diploid cells, P-M-A-T + cytokinesis; for growth and repair.
- Meiosis: germ cells, 4 haploid cells, 2 divisions, crossing over, variation.
- Scurvy = Vit C; Rickets = Vit D; Night blindness = Vit A; Goitre = iodine; Beriberi = Vit B1; Kwashiorkor = protein.

Unit IV

- Prokaryotic labels: cell wall, capsule, plasma membrane, nucleoid, plasmid, ribosomes (70S), mesosome, flagellum, pili.
- Mitochondria = powerhouse; lysosome = suicide bag; chloroplast = photosynthesis.
- Permanent tissue: Simple (parenchyma, collenchyma, sclerenchyma); Complex (xylem = water, phloem = food).

Unit V

- Four animal tissues: epithelial, connective, muscular, nervous.
- Neuron: cyton (nucleus + Nissl), dendrites (receive), axon (carry away), myelin, nodes of Ranvier, synapse. Types: sensory, motor, interneuron.
- NCF-2005 (NCERT, Yash Pal): connect knowledge to life, no rote, constructivism, activity-based. Six validities: cognitive, content, process, historical, environmental, ethical.

Nutrition modes (Unit V)

- Autotrophic (plants) vs heterotrophic (animals).
- Heterotrophic: holozoic, saprophytic, parasitic.
- Feeding: filter, fluid, bulk.

For more practice and bigger pictures

Diagram *Reference* Links

Reliable pages that show each structure-and-function diagram in detail. Open them to study the full-colour version or to copy a diagram for practice.

Neuron · structure and function (Q9)

BYJU'S — Diagram of Neuron with labels
byjus.com/biology/diagram-of-neuron/

GeeksforGeeks — Neuron diagram, structure and function
geeksforgeeks.org/biology/neuron-diagram-structure-function/

Vedantu — Neuron diagram: structure, parts, functions, types
vedantu.com/biology/neuron-diagram

Prokaryotic (bacterial) cell (Q7)

GeeksforGeeks — Diagram of prokaryotic cell
geeksforgeeks.org/biology/prokaryotic-cell-diagram/

BYJU'S — Prokaryotic cell, definition and structure
byjus.com/biology/prokaryotic-cells/

Wikipedia — Prokaryote cell diagram (free SVG) en.wikipedia.org/wiki/File:Prokaryote_cell_diagram.svg

Mitosis · stages (Q5)

GeeksforGeeks — Labelled diagram of mitosis
geeksforgeeks.org/biology/diagram-of-mitosis/

BYJU'S — Mitosis: stages and diagram
byjus.com/biology/mitosis/

Vedantu — Mitosis explained: stages and diagrams
vedantu.com/biology/mitosis

Permanent tissues · plant (Q8)

GeeksforGeeks — Permanent tissues, diagram and types
geeksforgeeks.org/permanent-tissues-class-9-biology/

Vedantu — Permanent tissue: types, structure, functions
vedantu.com/biology/permanent-tissue

Toppr — Plant tissues: xylem and phloem
toppr.com/guides/biology/anatomy-of-flowering-plants/plant-tissues/

Animal tissues · epithelial, connective, muscular, nervous (Unit V)

GeeksforGeeks — Animal tissue: structure and types
geeksforgeeks.org/biology/animal-tissues-types-structure/

BYJU'S — Types of animal tissue
byjus.com/biology/animal-tissue-types/

Vedantu — Types of animal tissue: diagram and functions
vedantu.com/biology/types-of-animal-tissue

Cell organelles · plant and animal cell (Unit IV)

BYJU'S — Plant cell: structure, function, diagram
byjus.com/biology/plant-cell/

BYJU'S — Well-labelled diagram of animal cell
byjus.com/biology/diagram-of-animal-cell/

Vedantu — Animal cell diagram with parts and functions
vedantu.com/biology/well-labelled-diagram-of-animal-cell

Balanced diet · components (Q6)

BYJU'S — Balanced diet: nutrients and deficiency diseases

byjus.com/cbse/balanced-diet-deficiency-diseases/

Toppr — Balanced diet: components and importance
toppr.com/guides/science/components-of-foods/balanced-diet/

Vedantu — Components of food: chart and functions
vedantu.com/biology/components-of-food

✦

Prepared from the BNMU Madhepura 2-Year B.Ed syllabus (pages 51–52), the 2025 PSS-5 question paper, and the "Excellent B.Ed Guide & Guess" reference pages. Diagrams drawn in-house; links are to open educational pages for further study. Visual style adapted from the Soft Editorial template system. Write your answers in your own words.